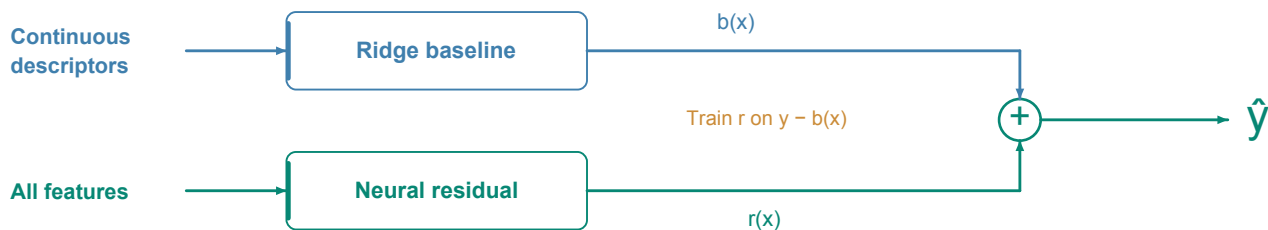


a Ridge + neural residual

Combine a continuous baseline with a learned correction.



Size-held-out MAE 1.391 · No tree-leaf ceiling in the correction

A linear trend is not a validated asymptotic free-energy law.

b Size-sensitive message passing

Count and aggregate atom states, not only fragment presence.

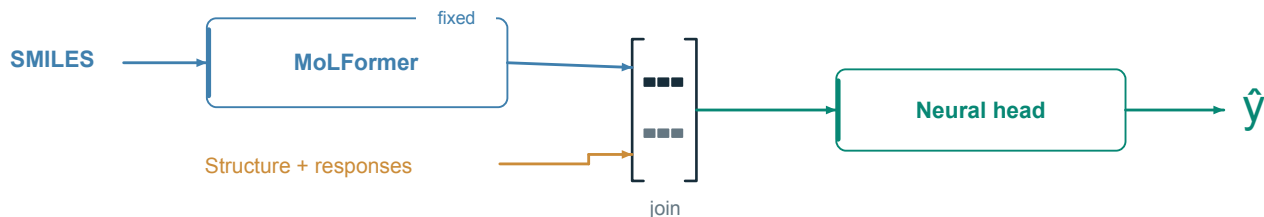


External MAE 1.168 · Without responses: 1.311

Shown: sum / mean / count readout; an atomic-plus-global variant was also tested.

c Frozen MolFormer + neural head

Reuse a pretrained sequence representation alongside response information.



External MAE 1.111 · Responses help in all three cohorts

The encoder stays fixed; the endpoint learns from hydration labels.